Lecture 15 - Exam 2 Review

(16) prove

$$\frac{2^{n+1}-1}{n+1} \quad 1 + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \dots$$

Solution:

$$(1+x)^n = \binom{n}{0} + 1\binom{n}{1} + \dots + \binom{n}{n}x^n$$

$$= \sum_{k=0}^{n}\binom{n}{k}x^k$$

Integrate both sides w.r.t. x

$$\int_0^1 (1+x)^n\,dx = \sum_{k=0}^{n}\binom{n}{k}\int_0^1 x^k\,dx \longrightarrow$$

important: bounds can be arbitrary, but $x=1$ gives the desired form and $x=0$ is easy to cancel out

$$\frac{(1+x)^{n+1}}{n+1}\Big|_0^1 = \sum_{k=0}^{n}\binom{n}{k}\left(\frac{x^{k+1}}{k+1}\Big|_0^1\right)$$

$$\frac{2^{n+1}}{n+1} - \frac{1^{n+1}}{n+1} = \sum_{k=0}^{n}\binom{n}{k}\left(\frac{1^{k+1}}{k+1} - \frac{0^{k+1}}{k+1}\right)$$

$$\frac{2^{n+1}-1}{n+1} = \sum_{k=0}^{n}\binom{n}{k}\frac{1}{k+1}$$

□

Remark: cannot use indefinite integral!

consider:

$$\frac{(1+x)^{n+1}}{n+1} + C_1 = \sum_{k=0}^{n}\binom{n}{k}\left(\frac{x^{k+1}}{k+1} + C_2\right)$$

finding C_1 and C_2 might be difficult!

(26) prove

$$\sum_{k=0}^{n} \binom{n}{k}\binom{n}{k-1} = \binom{2n+1}{n+1} - \binom{2n}{n}$$

~~Proof~~

Scratch work

consider $\binom{2n+1}{n+1}$ ← special element

choose $n+1$ people from $2n+1$

~~from $2n+1$~~

$\binom{2n}{n}$ ← choose n people from $2n$ total

Thus by subtraction principle, gives # ways to choose group without special element

Now consider

$$\sum_{k=1}^{n} \binom{n}{k}\binom{n}{n+1-k}$$ → select $n+1$ people from 2 distinct groups, one always picked from woman

Proof

consider the # ways to form a committee of $n+1$ people from a group of $n+1$ women and n men. Consider a special women called apple, let count how many # a committee with apple can be formed.

$\binom{2n}{n}$ → # ways to select committee with special person

$\binom{2n+1}{n+1}$ → # ways choose committee

Thus by subtraction principle,

$\binom{2n+1}{n+1} - \binom{2n}{n}$ → # ways to select committee without special person,

Alternatively, we can count the same # of ways by first picking k women, then picking $(n-k+1)$, so the total # of people in the committee is $n+1$, all over $k=1,2,\dots,n$. By addition principle:

$$\sum_{k=1}^{n} \underbrace{\binom{n}{k}}_{k \text{ women}} \underbrace{\binom{n}{n-k+1}}_{n-k+1 \text{ men}} \rightarrow \#\text{ways to choose } n+1 \text{ people without special person.}$$

Consequently, both count the same thing, so they must be equivalent ☐

(20) Find $a, b, c \in \mathbb{Z}$ such that

$$m^3 = a\binom{m}{3} + b\binom{m}{2} + c\binom{m}{1}$$

for all m. Then sum the series $1^3 + 2^3 + \dots + n^3$

Solution:

$$\binom{m}{3}\binom{3}{3}\binom{m}{1} = a\binom{m}{3} + b\binom{m}{2} + c\binom{m}{1}$$

choose 3 elements, repeats we fixed

order matters!

$$T = \{(x,y,z) \mid x,y,z \in \{1,\dots,m\}\}$$

$$m^3 = |T| = \binom{m}{3}3! + \binom{m}{2}\cdot\binom{2}{1}\cdot\binom{3}{1} + \binom{m}{1}1$$

choose 3 distinct and permute them

choose 2 distinct, choose which is unique, then determine which value is given to unique

choose 1 for all, one way

$$a = 3!,\ b = \binom{3}{2}\binom{3}{1},\ c = 1$$

Now

$$1^3+2^3+\dots+n^3=\sum_{k=1}^{n}k^3$$

Looks finish!

(48) Prove that for $n=m+1$ elements, either $m+1$ chain or $m+1$ antichain, $n,m\in\mathbb{Z}^+$

Proof Let $P=(X,\leq)$ be the poset with $|X|=n\cdot m+1$,

Suppose P has neither a chain of size $m+1$ nor an antichain of size $m+1$, Then the largest chain in P has size $\leq n$.

By Dilworth Theorem, X can be partitioned into $\leq n$ chains, say $C_1,\dots,C_r$, when $r\leq n$. Note that $|C_i|\leq m$ for all i.

Then, $|X|=|C_1|+\dots+|C_r|$

$$|X|=\underbrace{m+\dots+m}_{r}\leq m\cdot r$$

However, it is known that $|X|=n\cdot m+1$, so $|X|\neq m\cdot r$, $r\leq n$.

Therefore, by contradiction, at least one such chain or antichain must exist.